# Logical equivalence

**Logical equivalence** : Two compound propositions are **logically equivalent** means that they have the same truth values for all settings of truth values to their propositional variables.

**Tautology**: A compound proposition that evaluates to true for all settings of truth values to its propositional variables; it is abbreviated $T$.

**Contradiction**: A compound proposition that evaluates to false for all settings of truth values to its propositional variables; it is abbreviated $F$.

**Contingency**: A compound proposition that is neither a tautology nor a contradiction.

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